Task 5: Statistical Analysis of Iris Dataset

Objective

The objective of this task is to perform statistical analysis on the Iris Dataset using Python and SciPy. The analysis includes calculation of Mean, Median, Mode, P-Value, and implementation of a T-Test to validate data quantitatively.

Tools Used

- Python
- Pandas
- NumPy
- SciPy
- Matplotlib

Dataset

- Iris Dataset
- Total Records: 150
- Features:
  - Sepal Length
  - Sepal Width
  - Petal Length
  - Petal Width
  - Species

Statistical Analysis

Mean

The mean of Sepal Length was calculated.

Mean = 5.8433

Median

The median of Sepal Length was calculated.

Median = 5.8

Mode

The most frequently occurring value of Sepal Length was calculated.

Mode = 5.0

T-Test and P-Value Analysis

A two-sample T-Test was performed to compare the Sepal Length of Iris-setosa and Iris-versicolor species.

T-Statistic = -10.521
P-Value = 8.985×10^{-18}

Interpretation
The obtained P-Value is significantly smaller than the significance level of 0.05.

Since:
P-Value < 0.05

The Null Hypothesis is rejected.

This indicates that there is a statistically significant difference between the Sepal Length of Iris-setosa and Iris-versicolor species.

Visualization
A histogram was created to visualize the distribution of Sepal Length values in the Iris Dataset.

Conclusion
The Iris Dataset was successfully analyzed using statistical techniques. Mean, Median, and Mode were calculated, and a T-Test was conducted to compare species. The very small P-Value confirms a statistically significant difference between the selected groups. Therefore, the dataset was quantitatively validated through statistical analysis.

Outcome
The task successfully demonstrated the use of Python, Pandas, NumPy, and SciPy for statistical analysis and hypothesis testing on real-world data.